

Atmospheric Dynamics

Homework 1

Complete coursework manuscript

Problem 1

This question considers how much energy you emit.

- (a) Compute your total energy output in Watts. Assume a body temperature of 37°C and a surface area of 1.8 m^2 .
- (b) Compute how much energy you emit in a day, first in Joules, then convert to food Calories. Recall:

$$1 \text{ Calorie} = 1000 \text{ calories} = 4184 \text{ Joules}.$$

- (c) Convert this energy into the number of 16 oz steaks (each providing 1200 food Calories), and compute the monetary cost assuming beef is \$25 per pound. Compare with oats costing \$3 per pound for 1800 Calories. Compute the CO_2 footprint using:

$$\text{Beef: } 70 \text{ kg CO}_2/\text{pound}, \quad \text{Oats: } 0.5 \text{ kg CO}_2/\text{pound}.$$

Express the annual CO_2 emissions in terms of round trips to Vienna, assuming each trip generates 2.5 metric tons CO_2 .

- (d) Explain why this calculation overestimates human energy needs.

Solution to Problem 1

(a) Total power output in Watts

The body temperature in Kelvin is

$$T = 37^{\circ}\text{C} = 310.15 \text{ K}.$$

and the surface area is

$$A = 1.8 \text{ m}^2.$$

According to the Stefan–Boltzmann law, the radiative flux per area is

$$I = \sigma T^4,$$

where the Stefan–Boltzmann constant is

$$\sigma = 5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}.$$

The total emitted power is

$$P = IA = \sigma AT^4.$$

By the formula,

$$P = (5.67 \times 10^{-8})(1.8)(310.15^4) \text{ W},$$

we obtain

$$P \approx 944.37 \text{ W}.$$

Thus, the total energy output

$$\boxed{P \approx 944.37 \text{ W}}.$$

(b) Energy emitted a day

A day contains

$$t = 86400 \text{ s.}$$

Hence, the total emitted energy is

$$E = Pt \approx 944.37 \times 86400 \approx 8.16 \times 10^7 \text{ J.}$$

Convert Joules to food Calories (we denote Cal for food Calories):

$$1 \text{ Cal} = 4184 \text{ J.}$$

Therefore,

$$\text{Calories} = \frac{8.1 \times 10^7}{4184} \approx 1.9 \times 10^4 \text{ Cal.}$$

Thus, the energy emitted a day is

$$E \approx 8.16 \times 10^7 \text{ J/day} \quad \text{and} \quad \approx 1.9 \times 10^4 \text{ Cal/day.}$$

(c) Equivalent number of steaks and cost

Each 16 oz steak provides 1200 Cal. Thus the number of steaks required is

$$N_{\text{steak}} = \frac{1.9 \times 10^4}{1200} \approx 16.$$

At \$25 per pound (16 oz), the daily cost is

$$C_{\text{day}} = 16 \times 25 = 400 \text{ USD/day.}$$

For oats, providing 1800 Cal/lb at \$3 per pound, the amount needed is

$$N_{\text{oats}} = \frac{1.9 \times 10^4}{1800} \approx 10.6 \text{ lb/day,}$$

at cost:

$$C_{\text{oats}} \approx 10.6 \times 3 = 32 \text{ USD/day.}$$

For CO₂ emissions:

$$\text{Beef: } 70 \text{ kg CO}_2/\text{lb}, \quad \text{Oats: } 0.5 \text{ kg CO}_2/\text{lb.}$$

Daily emissions:

$$E_{\text{beef}} \approx 16 \times 70 = 1120 \text{ kg CO}_2/\text{day,}$$

$$E_{\text{oats}} \approx 10.6 \times 0.5 \approx 5.3 \text{ kg CO}_2/\text{day.}$$

A year has 365 days. Annual emissions:

$$E_{\text{beef,yr}} \approx 1120 \times 365 \approx 4.1 \times 10^5 \text{ kg} = 410 \text{ metric tons,}$$

$$E_{\text{oats,yr}} \approx 5.3 \times 365 \approx 1.9 \times 10^3 \text{ kg} = 1.9 \text{ metric tons.}$$

Given one round-trip flight to Vienna produces 2.5 t CO₂:

$$\text{Beef strategy} \approx \frac{410}{2.5} \approx 160 \text{ round trips/year,}$$

$$\text{Oats strategy} \approx \frac{1.9}{2.5} \approx 0.75 \text{ round trips/year.}$$

(d) Why the estimate is unrealistic

This estimate drastically overstates true metabolic needs because a human is not an isolated blackbody in vacuum:

- The environment radiates energy back toward human body.
- Convection, evaporation, and conduction also dominate heat exchange.
- Metabolism is controlled by biochemical efficiency, not radiative equilibrium.

Hence, humans require only about 2000 Cal/day, not the ~ 19000 Cal/day as we predict.

Problem 2

The Voyager 1 probe receives decreasing solar radiation.

- (a) Assuming the probe is a sphere of albedo 0.1, determine its temperature as a function of distance r from the Sun.
- (b) Compute its current temperature using the current distance listed on NASA's website.
- (c) Find the distance required for the probe to approach the background temperature $T = 2.7$ K, ignoring other stars and CMB.

Solution to Problem 2

We model Voyager 1 as a sphere with albedo $\alpha = 0.1$. The solar irradiance at r follows the inverse-square law:

$$S(r) = S_0 \left(\frac{1 \text{ AU}}{r} \right)^2,$$

where $S_0 = 1361 \text{ W m}^{-2}$ is the solar constant at 1 AU. The inverse-square law follows from the property

$$S(1 \text{ AU}) = \frac{P}{4\pi(1 \text{ AU})^2}, S(r) = \frac{P}{4\pi r^2},$$

where P denotes the total power of the sun.

Let R be the probe radius. The absorbed solar power equals the emitted power:

$$(1 - \alpha) S(r) (\pi R^2) = (4\pi R^2) \sigma T^4,$$

where $\sigma = 5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$ is the Stefan-Boltzmann constant.

Canceling πR^2 yields:

$$\frac{(1 - \alpha) S(r)}{4} = \sigma T^4.$$

(a) Temperature as a function of r

Substitute $S(r) = S_0(1 \text{ AU}/r)^2$ into the balance:

$$\frac{(1 - \alpha) S_0}{4} \left(\frac{1 \text{ AU}}{r} \right)^2 = \sigma T(r)^4.$$

Solve for $T(r)$:

$$T(r) = \left(\frac{(1 - \alpha)S_0}{4\sigma} \right)^{1/4} \left(\frac{1 \text{ AU}}{r} \right)^{1/2}.$$

In particular,

$$T(r) \propto r^{-1/2}.$$

(b) Current temperature

Let r_{now} be Voyager 1's current distance from the Sun. Then,

$$T_{\text{now}} = \left(\frac{(1 - \alpha)S_0}{4\sigma} \right)^{1/4} r_{\text{now}}^{-1/2},$$

where r_{now} is dimensionless since it is in astronomical units (AU).

Using $\alpha = 0.1$, $S_0 = 1361 \text{ W m}^{-2}$, and $\sigma = 5.67 \times 10^{-8}$ gives

$$T_{\text{now}} \approx \frac{269}{\sqrt{r_{\text{now}}}} \text{ K}.$$

We check the r_{now} on NASA

$$r_{\text{now}} = 169.57 \text{ AU}.$$

Thus,

$$T_{\text{now}} \approx 20.7 \text{ K}$$

(c) How far does it need to reach $T = 2.7 \text{ K}$

Set $T(r) = 2.7 \text{ K}$:

$$2.7 = \left(\frac{(1 - \alpha)S_0}{4\sigma} \right)^{1/4} \left(\frac{1}{\sqrt{r}} \right).$$

Thus,

$$r = \left(\frac{(1 - \alpha)S_0}{4\sigma} \right)^{1/2} \left(\frac{1}{2.7} \right)^2 \text{ AU},$$

$$r \approx 9.9 \times 10^3 \text{ AU}.$$

SO, voyager needs to travel $9.9 \times 10^3 \text{ AU}$ from the Sun to approach 2.7 K .

Problem 3

Consider Earth as a leaky greenhouse.

- (a) Find surface temperature T_s and atmospheric temperature T_a assuming:

$$S_0 = 1361 \text{ W/m}^2, \quad \alpha = 0.300, \quad \epsilon = 0.700.$$

- (b) Doubling CO_2 corresponds to an additional 4 W/m^2 of heating at the surface. Find the new ϵ that produces this additional downward radiation.
- (c) Find the change in surface temperature required to restore equilibrium.
- (d) Assume emissivity increases by 0.01 per Kelvin:

$$\epsilon(T_s) = \epsilon_0 + 0.01 (T_s - T_{s,0}).$$

Compute the new equilibrium temperature after warming.

Solution to Problem 3

(a) Find surface and atmospheric temperature

Atmospheric balance.

$$\epsilon\sigma T_s^4 = 2\epsilon\sigma T_a^4 \implies T_a^4 = \frac{1}{2} T_s^4.$$

Surface balance.

$$\frac{(1-\alpha)S_0}{4} + \epsilon\sigma T_a^4 = \sigma T_s^4.$$

Substitute $T_a^4 = T_s^4/2$:

$$\begin{aligned} \frac{(1-\alpha)S_0}{4} &= \sigma T_s^4 \left(1 - \frac{\epsilon}{2}\right), \\ T_s &= \left(\frac{S_0(1-\alpha)}{4\sigma} \cdot \frac{1}{1-\epsilon/2}\right)^{1/4}. \end{aligned}$$

Thus,

$$T_s \approx 283.53 \text{ K}, \quad T_a = \frac{T_s}{2^{1/4}} \approx 238.42 \text{ K}.$$

(b) Find the new value of ϵ

A CO₂ doubling produces additional heating

$$(\epsilon_1 - \epsilon_0)\sigma T_a^4 = 4.$$

Hence,

$$\epsilon_1 = \epsilon_0 + \frac{4}{\sigma T_a^4} \approx 0.722.$$

(c) Find the change in surface temperature

New equilibrium surface temperature satisfies

$$T_{s,1}^4 = \frac{(1-\alpha)S_0}{4\sigma(1-\epsilon_1/2)}.$$

Thus,

$$T_{s,1} \approx 284.73 \text{ K}, \quad \Delta T_s = T_{s,1} - T_{s,0} \approx 1.20 \text{ K},$$

where $T_{s,0} = T_s$.

(d) Temperature-dependent emissivity

Assume

$$\epsilon(T_s) = \epsilon_1 + 0.01 (T_{s,eq} - T_{s,0}).$$

The equilibrium condition becomes

$$T_{s,eq}^4 = \frac{F_\odot}{\sigma \left(1 - \frac{1}{2}\epsilon(T_s)\right)} = \frac{F_\odot}{\sigma \left(1 - \frac{1}{2}[\epsilon_1 + 0.01 (T_{s,eq} - T_{s,0})]\right)}.$$

Solving numerically gives

$$T_{s,eq} \approx 286.29 \text{ K}.$$

Total warming:

$$\Delta T_s \approx 286.29 - 283.53 \approx 2.76 \text{ K}.$$

Problem 4

Highly opaque atmospheres can be modeled using multiple absorbing layers.

- (a) Set up an n -layer opaque ($\epsilon = 1$) greenhouse model and solve for temperatures of each layer and the surface as a function of emission temperature T_e .

- (b) Using:

$$S_0 = 2632 \text{ W/m}^2, \quad \alpha = 0.77,$$

determine how many layers are required to obtain a surface temperature of 760 K (Venus).

- (c) Estimate the number of “atmospheres” of water vapor available in Earth’s oceans (ratio of water mass to atmospheric mass). Use ocean area and mean depth for a back-of-the-envelope estimate.

Solution to Problem 4

We consider an atmosphere consisting of n opaque ($\epsilon = 1$) infrared-absorbing layers. Let T_s be the surface temperature, and T_i ($i = 1, \dots, n$) the temperature of the i -th layer. Let T_e be the planet’s emission temperature and we define $x_i := T_i^4$, $x_s := T_s^4$, and $x_e := T_e^4$.

(a) Energy balance with n opaque layers

Top boundary. The top layer radiates to space:

$$x_n = x_e.$$

Interior layers. For $1 < i < n$, absorption from above and below equals emitted radiation:

$$x_{i-1} + x_{i+1} = 2x_i.$$

Thus x_i is linear in i . Using $x_n = x_e$ and the top balance

$$x_{n-1} = 2x_n = 2x_e,$$

we obtain the full sequence:

$$\boxed{T_i^4 = (n - i + 1) T_e^4, \quad i = 1, \dots, n.}$$

Surface balance. Surface absorbs the solar flux σT_e^4 , and the downward longwave from layer 1, which satisfies:

$$x_s = x_e + x_1 = x_e + nx_e = (n + 1)x_e.$$

Hence,

$$\boxed{T_s^4 = (n + 1) T_e^4, \quad T_s = (n + 1)^{1/4} T_e.}$$

How many layers are needed

Given $S_0 = 2632 \text{ W/m}^2$ and albedo $\alpha = 0.77$, the absorbed solar flux is

$$\sigma T_e^4 = \frac{(1 - \alpha)S_0}{4} = \frac{0.23 \times 2632}{4} = 151.34 \text{ W/m}^2.$$

Thus,

$$T_e = \left(\frac{151.34}{\sigma} \right)^{1/4} \approx 227.29 \text{ K}.$$

We want a surface temperature of $T_s = 760 \text{ K}$. Using

$$T_s = (n + 1)^{1/4} T_e,$$
$$n + 1 = \left(\frac{T_s}{T_e} \right)^4 = \left(\frac{760}{227.29} \right)^4 \approx 125.00.$$

Therefore

$n \approx 124$ opaque layers are required for Venus.

(c) The number of “atmospheres” of water in Earth’s oceans?

We estimate the mass of Earth’s oceans and divide by the mass of the atmosphere.

Ocean mass. Approximate ocean area:

$$A_{\text{oc}} \approx 0.71 \cdot 4\pi R^2 \approx 3.62 \times 10^{14} \text{ m}^2,$$

with mean depth $d \approx 3700 \text{ m}$ and water density $\rho = 1000 \text{ kg/m}^3$:

$$M_{\text{oc}} \approx \rho A_{\text{oc}} d \approx 1000 \times 3.62 \times 10^{14} \times 3700 \approx 1.34 \times 10^{21} \text{ kg}.$$

Atmosphere mass.

$$M_{\text{atm}} \approx 5.15 \times 10^{18} \text{ kg}.$$

Mass ratio.

$$\frac{M_{\text{oc}}}{M_{\text{atm}}} \approx \frac{1.34 \times 10^{21}}{5.15 \times 10^{18}} \approx 2.6 \times 10^2.$$

Therefore,

Earth’s oceans contain roughly 260 times the mass of Earth’s atmosphere.
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